

## How it works

This document describes the logic inside `Audit committee dashboard.html` so a reviewer can follow what the tool does without reading the code.

## What the tool does

`Audit committee dashboard.html` is a single-page, browser-based dashboard builder. You upload a spreadsheet (`.xlsx`, `.xls` or `.csv`), and it turns the rows into a set of summary figures and charts without any set-up. It was built and tested around audit issue logs, but it works on any table with a header row.

In short, the tool:

- reads the file and works out what each column holds (dates, numbers, risk ratings, statuses, processes or functions, plain text);
- shows KPI cards, such as total rows, % open and High / Critical count, wherever the data supports them;
- picks four starter charts (bar, line, pie and doughnut) from the columns it found;
- lets you edit, add or remove charts, and correct any column type it guessed wrongly;
- exports the results to Excel or PowerPoint as summary numbers only.

Everything runs in the browser, so the uploaded data never leaves your machine.

## 1. Pipeline

```
Upload file -> Parse (SheetJS) -> Detect column types -> Pick starter charts  
-> Aggregate -> Render (Chart.js) -> Edit / add / remove -> Export
```

## 2. Reading the file

- Accepts `.xlsx`, `.xls`, `.csv`. Anything else shows an error message.
- The first non-empty row is the header row. Blank rows are skipped.
- Blank header cells become `Column N`; duplicate headers get a numeric suffix.
- A sheet with no data rows, or an empty sheet, shows a clear error.
- Multi-sheet workbooks show a sheet selector.

## 3. Column type detection

Each column is classified from up to 200 non-blank values, using the header name and the values together. Rules are checked in this order:

| Order | Type               | Rule                                                                                                                                                                                                     |
|-------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <b>Date</b>        | at least 90% of values parse as dates, or the header contains <i>date</i> , <i>time</i> , <i>created</i> , <i>due</i> , <i>opened</i> , <i>raised</i> and all values look like Excel date serials        |
| 2     | <b>Number</b>      | at least 90% of values parse as numbers (commas, %, currency symbols are stripped)                                                                                                                       |
| 3     | <b>Risk Rating</b> | header contains <i>risk</i> , <i>rating</i> , <i>severity</i> , <i>priority</i> with 15 or fewer distinct values, or at least 60% of values are recognised rating words with 10 or fewer distinct values |

| Order | Type                    | Rule                                                                                                                  |
|-------|-------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 4     | <b>Status</b>           | header contains <i>status</i> with 10 or fewer distinct values, or at least 60% of values are recognised status words |
| 5     | <b>Process/Function</b> | header contains <i>process, function, department, business unit, area, domain</i> with more than one distinct value   |
| 6     | <b>Text</b>             | everything else                                                                                                       |

Recognised rating words: critical, very high, high, medium, moderate, med, low, very low. Recognised status words: open, closed, in progress, overdue, resolved, pending, done, complete(d), new, not started, on hold, cancelled.

**Dates:** ISO (2026-03-31), dd/mm/yyyy or mm/dd/yyyy, Mon yyyy, dd-Mon-yyyy and Excel serials are supported. Day-first is the default; if any value in the column has a first part above 12 it is treated as day-first, and if a second part is above 12, month-first.

Any wrong guess can be fixed in the **Detected columns** panel; charts and KPIs recompute immediately.

## 4. KPI cards

| KPI             | Shown when                  | Calculation                                                                                                        |
|-----------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------|
| Total rows      | always                      | count of data rows                                                                                                 |
| Columns         | always                      | count of columns                                                                                                   |
| % Open          | a Status column exists      | share of non-blank statuses that are not <i>closed, resolved, done, complete, completed, remediated, cancelled</i> |
| High / Critical | a Risk Rating column exists | rows rated High or Critical (after mapping <i>very high</i> to Critical)                                           |

## 5. Starter charts

| Slot | Chart    | Dimension used                             | Fallback if missing                                                  |
|------|----------|--------------------------------------------|----------------------------------------------------------------------|
| 1    | Bar      | first Process/Function column              | first text column with 2 to 500 distinct values, then Risk or Status |
| 2    | Line     | first Date column, grouped by month        | average of the first Number column by row order, then an empty card  |
| 3    | Pie      | first Risk Rating column                   | first Status column, then a low-cardinality text column              |
| 4    | Doughnut | first Status column (other than the pie's) | a low-cardinality text column                                        |

When a fallback is used, the chart subtitle says so, for example *No Process/Function column, using "Department"*.

## 6. Aggregation

- Metric can be **count of rows**, **sum** or **average** of a number column.

- Dates are bucketed by calendar month; undated rows appear as (blank).
- Rating columns are sorted Critical to Low and drawn in fixed colours (Critical dark red, High orange, Medium amber, Low green).
- Other categories are sorted by value; beyond **Top N** (default 10, range 2 to 50) the remainder is grouped as **Other**.
- Doughnut charts show the total in the centre (not shown for averages).

## 7. Editing charts

- Pencil icon: change title, chart type, dimension, metric, metric column, Top N.
- Cross icon: remove a chart, with an **Undo** option in the toast message.
- + **Add Chart**: create any additional chart from the dialog.
- Dialog is keyboard accessible (focus trap, Esc to close).

## 8. Export

| Format     | Content                                                                                       |
|------------|-----------------------------------------------------------------------------------------------|
| Excel      | Summary sheet (source file, rows, columns, KPIs) plus one sheet per chart with its data table |
| PowerPoint | Title slide, key figures slide, then one native chart per slide                               |

Exports contain summary numbers only, capped at 50 points per chart (the most recent 50 for line charts). The raw uploaded data is not included.

## 9. Privacy and dependencies

No uploaded data leaves the browser. External requests are limited to loading these libraries:

| Library   | Version | Used for                                                           |
|-----------|---------|--------------------------------------------------------------------|
| SheetJS   | 0.18.5  | reading and writing Excel                                          |
| Chart.js  | 4.4.1   | on-screen charts                                                   |
| PptxGenJS | 3.12.0  | PowerPoint export (loaded on demand from jsDelivr, cdnjs or unpkg) |

## 10. Manual test cases

| # | Test                                                                                       | Expected                                                     |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 1 | Upload <code>audit_issues_sample.csv</code>                                                | 60 rows, 10 columns, 45% open, 18 High/Critical, four charts |
| 2 | Upload <code>audit_issues_sample.xlsx</code> , switch to <code>Control_Test_Results</code> | 22 rows, no Status or Risk KPIs, charts adapt                |
| 3 | Upload a <code>.txt</code> file                                                            | "Unsupported file type" message                              |
| 4 | Upload an empty workbook                                                                   | Clear error, no crash                                        |

| # | Test                                                                   | Expected                                           |
|---|------------------------------------------------------------------------|----------------------------------------------------|
| 5 | Change a column type in Detected columns                               | Charts and KPIs recompute                          |
| 6 | Remove a chart then click Undo                                         | Chart returns in its original position             |
| 7 | Add a chart: Sum of Exceptions by Process (Control_Test_Results sheet) | Hire to Retire and IT General Controls show 6 each |
| 8 | Save as Excel and as PowerPoint                                        | Files download with one sheet or slide per chart   |
| 9 | Switch the OS to dark mode                                             | Colours adapt, charts redraw                       |